

Week 2 - ZKP

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Session 1: Zero-Knowledge Proof Basics

KZG Commitments

Imagine you bake a chocolate cake and place it inside a sealed box. Later, you want to convince your friend that the whole cake is indeed chocolate without letting them eat it all. You cut a tiny slice and hand it over, and they check that it tastes like chocolate.

KZG commitments are a way to commit to a polynomial so that later one can prove its value at a certain point without revealing the entire polynomial. This idea is important because it allows efficient verification of complex computations while keeping the data hidden. KZG commitments are used in modern proof systems such as PLONK. They are powerful because the proof is very small and quick to verify, but they require a trusted setup process and are not resistant to quantum attacks.

Basics of ZKPs

A Zero-Knowledge Proof (ZKP) involves two roles: the prover, who wants to prove that a statement is true, and the verifier, who wants to be convinced of the truth. The prover uses a secret piece of information, called the witness, to construct the proof. In interactive proofs, the prover and verifier exchange several rounds of messages, while in non-interactive proofs the prover sends a single proof message, which is common in blockchain applications.

Properties of ZKPs

Zero-Knowledge Proofs have three essential properties. Completeness means that if the statement is true, an honest prover can always convince the verifier. Soundness means that if the statement is false, a cheating prover cannot trick the verifier into accepting. Finally, the zero-knowledge property ensures that the verifier learns nothing beyond the fact that the statement is true, protecting the prover's secret information.

Session 2: Advanced Zero-Knowledge Proof Systems

Types of ZKP: SNARKs and STARKs

ZK-SNARKs are a type of proof system that produce very small proofs and allow for fast verification. They are widely used in blockchains today, such as in Zcash, but usually require a trusted setup. ZK-STARKs, on the other hand, do not need a trusted setup and are considered transparent. The trade-off is that STARK proofs are larger, although they scale better for very large computations. In short, SNARKs are efficient and compact but rely on setup assumptions, while STARKs are larger but more secure and transparent.

VMs, EVM, and zkVM

A Virtual Machine (VM) is an environment that runs programs, such as Ethereum's Virtual Machine (EVM), which executes smart contracts. A zkVM is a virtual machine designed to produce a zero-knowledge proof that a program was executed correctly. This means that instead of re-running a program, one can verify its execution with just a proof. zkVMs are important in scaling blockchains, as they make it possible to prove large amounts of computation off-chain and only verify a small proof on-chain. This approach reduces costs, saves gas, and makes technologies such as zkRollups possible.

Key Takeaways

Zero-Knowledge Proofs let us prove knowledge without revealing the secret itself. KZG commitments provide the building blocks for efficient proof systems. SNARKs and STARKs represent the two main families of advanced ZKPs, each with different trade-offs. Finally, zkVMs connect these cryptographic techniques to real-world computation, enabling scalable and secure blockchain applications.